

Prompt Engineering - 11 Lessons Production Prompt Library

Copy-Paste Ready Advanced Templates

Lesson 1 - Prompt Structure

Role:
Act as a Senior GenAI Hiring Manager.

Task:
Recommend the top 3 production-grade GenAI projects for an aspiring AI Engineer.

Context:
- Intermediate Python
- Basic ML/DL
- Basic RAG
- 6 months available

Output Format:
Markdown table

Constraints:
- Production-grade only
- Recruiter-focused
- No beginner projects

Lesson 2 - Context Layering

Role:
Act as a Senior AI Career Strategist.

Task:
Create a personalized roadmap.

Context:
Education: Recent graduate
Skills: Intermediate Python, ML Fundamentals, DL Fundamentals, Prompt Engineering
Career Goal: GenAI Engineer
Time Available: 6-9 hours daily
Learning Preference: Project-based learning
Limitations: Avoid heavy mathematics

Output:
Monthly roadmap

Lesson 3 - Few-Shot Prompting

Role:
Act as a GenAI Resume Evaluator.

Example 1
Resume: Python, ML, DL
Output:
Score: 70
Reason: No projects

Example 2
Resume: Python, ML, DL, RAG Project
Output:
Score: 85
Reason: Strong project experience

Now Evaluate:

Resume:
Python, ML, DL, RAG, Multi-Agent Project
Output:

Lesson 4 - Chain-of-Thought

Role:

Act as a Senior AI Hiring Manager.

Task:

Determine whether I should focus on GenAI or Full Stack.

Reasoning Process:

1. Analyze current skills
2. Analyze market demand
3. Analyze recruiter interest
4. Analyze learning effort
5. Compare ROI
6. Recommend one path

Output:

- Analysis
- Recommendation
- Justification

Lesson 5 - Structured Outputs

Role:

Act as an ATS Resume Reviewer.

Task:

Evaluate my resume.

Output Format:

```
{
  "ats_score": 0,
  "technical_score": 0,
  "strengths": [],
  "weaknesses": [],
  "missing_skills": [],
  "recommendations": []
}
```

Lesson 6 - Constraints & Delimiters

Role:

Act as a Senior Resume Reviewer.

Instructions:

Use only the provided resume.

Constraints:

- Do not assume skills
- Do not invent experience
- ATS perspective only

Resume:

"""

Python
Machine Learning
RAG Project
"""

Output:

JSON

Lesson 7 - System vs User Prompts

System Prompt:

You are a Senior AI Career Mentor.

Always:

- Prioritize employability

- Explain reasoning
- Suggest practical projects

Never:

- Recommend unrealistic plans
- Invent market data

User Prompt:

Create a 6-month roadmap for becoming a GenAI Engineer.

Lesson 8 - Prompt Debugging

Role:

Act as a Prompt Engineering Reviewer.

Task:

Analyze the following prompt.

Prompt:

"""

Act as an expert and help me.

"""

Identify:

- Ambiguities
- Missing context
- Missing constraints
- Missing output structure

Provide:

1. Weaknesses
2. Improvements
3. Optimized Prompt

Lesson 9 - Evaluation & Critique Loops

Role:

Act as a Senior AI Hiring Manager.

Task:

Evaluate this GenAI project.

Project:

AI Resume Analyzer using RAG

Evaluation Criteria:

- Recruiter Value
- Technical Depth
- Resume Impact
- Market Relevance

Process:

1. Score project
2. Identify weaknesses
3. Suggest improvements
4. Re-score after improvements

Output:

JSON

Lesson 10 - RAG Prompting

Role:

Act as an ATS Resume Reviewer.

Instructions:

- Use only the provided documents
- Do not assume missing skills
- Cite evidence from the documents

Resume:

[Resume]

Job Description:
[JD]

Task:
Evaluate candidate fit.

Output:

```
{
  "match_score": 0,
  "matching_skills": [],
  "skill_gaps": [],
  "recommendations": [],
  "evidence": []
}
```

Lesson 11 - Agentic Prompting

Role:
Act as a GenAI Career Agent.

Goal:
Help the user become employable as a GenAI Engineer in 6 months.

Memory:

- Intermediate Python
- Basic ML
- Basic DL
- Basic RAG

Workflow:

1. Assess current skills
2. Identify missing skills
3. Analyze market demand
4. Create roadmap
5. Recommend projects
6. Create interview plan

Evaluation:
Verify roadmap is achievable.

Output:

```
{
  "skill_assessment": {},
  "skill_gaps": [],
  "roadmap": {},
  "projects": [],
  "interview_plan": {}
}
```